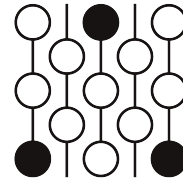


Review of Submission 4A

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

This submission makes some partial progress on the question. It divides the original problem into an ‘easy’ case and a ‘hard’ case and proves the easy case. It also proves a weaker estimate in the hard case.

Correctness. The proofs in this submission are correct. There is one place where there is either a small error or bad writing, but it can be fixed easily.

Novelty. I think there is some novelty in the submission. The paper uses tools and ideas from the paper “Area-expanding embeddings of rectangles,” which it cites. It also adapts some of the proof ideas from that paper to a somewhat different situation. In the original paper, the argument pushes forward chains from the domain to the range, and uses the isoperimetric inequality in the range. In this submission, the argument pulls back chains from the range to the domain and uses the isoperimetric inequality in the domain. This is the most novel ingredient in the submission. It also gathers together a few known results from the literature and combines the information from these known results.

This submission contains some of the ideas in the author’s solution, but not the most novel idea in the solution.

2 Recommendation

I don’t think that this submission fits cleanly into the four categories provided. It does not claim to solve the whole problem but only to make partial progress. The proofs here are correct and could be published with minor clarification.

The partial work here makes some progress towards answering the question, but on the other hand, the most novel and important idea in the solution is missing.

On this basis, I recommend ‘major revisions’, because some meaningful progress has been made, but it would still require major additional work to answer the question posed.

3 Submission with inline comments

Partial proof: the first alternative and the remaining hard estimate

Problem statement and interpretation

I write the small universal constant in the problem as κ , reserving k for k -dilation. Rectangles are oriented Euclidean products and degree 1 means relative degree 1 on $H_4(R, \partial R; \mathbb{Z}) \rightarrow H_4(S, \partial S; \mathbb{Z})$.

The current proof is complete through the first alternative and through a conditional reduction of the hard case. It is not a complete proof of the stated dichotomy, because the averaged tightening estimate (EAT) below remains unproved. The arguments are written for smooth transverse maps; piecewise smooth maps may be treated by the same current-slicing argument, or by smoothing on strata with an arbitrarily small C^1 -loss and then absorbing that loss into the universal constants.

Standard inputs

All constants are positive and universal. If $\text{Dil}_2(f) \leq 1$, then $\text{Dil}_3(f), \text{Dil}_4(f) \leq 1$: for singular values $\sigma_1 \geq \dots \geq \sigma_4$, the inequality $\sigma_1 \sigma_2 \leq 1$ implies $\sigma_2 \leq 1$ and $\sigma_1 \dots \sigma_j \leq \sigma_1 \sigma_2$ for $j \geq 2$.

We use Guth's map-of-pairs estimate [1, Estimate 1]. For $n = 4, k = 2, j = 1, l = 4$ it gives

$$R_1^2 \text{Vol}(R) \geq c_0 S_1^3 S_2 S_3 S_4. \quad (1)$$

We also use the small-volume relative isoperimetric estimate in rectangles [1, Theorem 3]: if Z is a relative 2-cycle in R with $A = \text{Mass } Z \leq c_* R_1 R_2$, then

$$\text{Fill}_R(Z) \leq C_* \max\{A^{3/2}, A^2/R_1\}.$$

Consequently

$$\text{Fill}_R(Z) \geq L \implies \text{Mass } Z \geq c \min\{R_1 R_2, L^{2/3}, (R_1 L)^{1/2}\}, \quad (2)$$

and, after decreasing c_* ,

$$\text{Mass } Z < c_* R_1 R_2 \implies \text{Fill}_R(Z) \leq C R_1 R_2^2. \quad (3)$$

The first alternative

Let Q be the concentric subrectangle of the (x_3, x_4) -face of S with side lengths $S_3/2, S_4/2$, and set

$$P_y = [0, S_1] \times [0, S_2] \times \{y\}, \quad y \in Q.$$

Lemma 1 (central target plane). *Every relative 3-chain in S with boundary $P_y, y \in Q$, has mass at least $c S_1 S_2 S_3$.*

Proof. Choose a 1-Lipschitz function ψ on the (x_3, x_4) -rectangle which vanishes on its boundary and satisfies $\psi \geq cS_3$ on Q . Put $\beta = \psi dx_1 \wedge dx_2$. Then $d\beta$ has comass at most 1, and Stokes' theorem gives

$$\text{Mass } Y \geq \left| \int_Y d\beta \right| = \left| \int_{P_y} \beta \right| \geq cS_1S_2S_3.$$

The relative boundary terms vanish because $dx_1 \wedge dx_2 = 0$ on the x_1, x_2 -faces and $\psi = 0$ on the x_3, x_4 -faces.

[Reviewer comment: The last justification is either a small error or bad writing, though it is easily fixed. The form $dx_1 \wedge dx_2$ vanishes on all faces except those in the x_1, x_2 direction, and ψ vanishes on those faces. Possibly it writes x_1 -face to mean a face where x_1 is constant, but this is unclear writing and not consistent with the convention above, where it says that Q is in the (x_3, x_4) face of S .] $\square$

Proposition 1. *There is $c_2 > 0$ such that, under the hypotheses of the problem, if*

$$R_1R_2^2 \leq c_2S_1S_2S_3,$$

then $R_3R_4 \geq c_2S_3S_4$.

Proof. Let $T = [[R]]$ be the fundamental relative current and put $F = (f_3, f_4)$. Since f has relative degree 1, $f_{\#}T = [[S]]$ as a relative 4-current. For a.e. $y \in Q$, slicing naturality gives

$$f_{\#}\langle T, F, y \rangle = \langle [[S]], (x_3, x_4), y \rangle = \pm P_y.$$

Write $Z_y = \langle T, F, y \rangle$. Any relative filling of Z_y in R pushes forward, using $\text{Dil}_3(f) \leq 1$, to a filling of P_y in S . Lemma 1 therefore gives

$$\text{Fill}_R(Z_y) \geq cS_1S_2S_3.$$

If c_2 is sufficiently small, (3) forces $\text{Mass } Z_y \geq c_*R_1R_2$. Since $J_2F \leq \text{Dil}_2(f) \leq 1$, the coarea formula yields

$$\text{Vol}(R) \geq \int_Q \text{Mass } Z_y dy \geq cR_1R_2S_3S_4.$$

Cancelling R_1R_2 proves the proposition. $\square$

The unconditional partial hard-case bound

Assume $R_1 = \alpha S_1 \leq S_1$ and that the first conclusion fails with a small constant. Proposition 1 gives

$$R_1R_2^2 \geq cS_1S_2S_3. \tag{4}$$

For the slices above, (2) with $L = cS_1S_2S_3$ gives

$$\text{Mass } Z_y \geq c \min\{R_1R_2, (S_1S_2S_3)^{2/3}, (R_1S_1S_2S_3)^{1/2}\}.$$

Using (4) and $R_1 = \alpha S_1$, every term in the minimum is at least $c\alpha^{1/2}S_1(S_2S_3)^{1/2}$. Hence

$$\text{Vol}(R) \geq c\alpha^{1/2}S_1S_2^{1/2}S_3^{3/2}S_4. \quad (5)$$

On the other hand, (1) gives

$$\text{Vol}(R) \geq c\alpha^{-2}S_1S_2S_3S_4. \quad (6)$$

Optimizing between (5) and (6) gives

$$\text{Vol}(R) \geq cS_1S_2^{3/5}S_3^{7/5}S_4.$$

Thus the remaining gap is exactly the high-aspect factor $(S_3/S_2)^{1/10}$.

A sufficient averaged hard-case estimate

Let $G = (f_1, f_2)$, keep $F = (f_3, f_4)$, and define

$$E_y = \int_{Z_y} \|dG|_{TZ_y}\|_{HS}^2 d\|Z_y\|, \quad I = \int_Q E_y dy, \quad q = |Q|.$$

At a regular point of F , let $K = \ker dF$, and let $a \geq b$ be the singular values of $dF|_{K^\perp}$; then $ab = J_2F$. For any unit $w \in K$, the mixed two-vector $df(w) \wedge df(v)$, where v is a unit vector with $|dF(v)| = a$, has a component of norm $|dG(w)|a$. Since $\text{Dil}_2(f) \leq 1$, $|dG(w)|a \leq 1$, and hence $|dG(w)|^2 J_2F \leq b/a \leq 1$. Summing over an orthonormal basis of K gives

$$\|dG|_{\ker dF}\|_{HS}^2 J_2F \leq 2.$$

By coarea,

$$I \leq C \text{Vol}(R). \quad (7)$$

Let $\Omega = \{y \in Q : E_y \leq 4I/q\}$, so $|\Omega| \geq 3q/4$.

For real normal currents define

$$\text{EFill}_f(y) = \inf\{\text{Mass } C : C \in N_3(R, \partial R; \mathbb{R}), f_\#(\partial C) = \pm P_y\}.$$

This class is nonempty by filling the actual slice Z_y . If C is admissible, then $f_\#C$ fills P_y and $\text{Mass}(f_\#C) \leq \text{Mass } C$; Lemma 1 gives

$$\text{EFill}_f(y) \geq cS_1S_2S_3. \quad (8)$$

Therefore

$$A_\Omega := \int_\Omega \text{EFill}_f(y) dy \geq cS_1S_2S_3q \geq cS_1S_2S_3^2S_4. \quad (9)$$

For $z \in Q$, let $\mathcal{B}_z$ be the set of $x \in R$ such that z belongs to the F -image of at least one coordinate 2-plaque through x . For fixed x , the set of such z 's has measure at most CR_3R_4 , because there are six plaques, each has area at most R_3R_4 , and $J_2F \leq 1$ on each plaque. Hence, for every normal 3-current B ,

$$\frac{1}{q} \int_Q \text{Mass}(B \llcorner \mathcal{B}_z) dz \leq C \frac{R_3R_4}{S_3S_4} \text{Mass } B. \quad (10)$$

The following estimate is sufficient for the desired theorem. Take measurable near-minimizers C_y^0 with $\text{Mass } C_y^0 \leq 2 \text{ EFill}_f(y)$; standard measurable selection in the separable flat topology, or equivalently outer integration with ε -minimizers, is enough for this formal step. One needs

$$A_\Omega \leq C \left(R_1 I + \frac{I^2}{S_1 q} \right) + \frac{C}{q} \int_Q \int_\Omega \text{Mass}(C_y^0 \lrcorner \mathcal{B}_z) dy dz. \quad (\text{EAT})$$

Assuming (EAT), (10) and $\text{Mass } C_y^0 \leq 2 \text{ EFill}_f(y)$ give

$$A_\Omega \leq C \left(R_1 I + \frac{I^2}{S_1 q} \right) + C \frac{R_3 R_4}{S_3 S_4} A_\Omega.$$

If $R_3 R_4 \leq \kappa S_3 S_4$, the last term is absorbed by taking κ small. If also $R_1 \leq \kappa S_1$ and $I \leq \delta S_1 S_2^{1/2} S_3^{3/2} S_4$, then, since $q \sim S_3 S_4$,

$$R_1 I \leq C \kappa \delta S_1 S_2 S_3^2 S_4, \quad \frac{I^2}{S_1 q} \leq C \delta^2 S_1 S_2 S_3^2 S_4.$$

For small δ, κ this contradicts (9). Thus $I \geq c S_1 S_2^{1/2} S_3^{3/2} S_4$, and (7) gives the desired second alternative.

Why the remaining estimate is non-routine

The estimate (EAT) cannot be obtained by a standard source-local cellular deformation in the quotient by f -null chains without a new idea. The obstruction is already visible in a two-cell model. Let A, A' be two oriented 2-cells with $f_\# A = f_\# A' = P$. Then $N = A - A' \in \ker f_\#$. If an exact-plaque bad set contains A but not A' , the source-local projection P_{bad} with $P_{\text{bad}} A = A$, $P_{\text{bad}} A' = 0$ gives $f_\# P_{\text{bad}} N = P \neq 0$. Dually, a quotient cochain has equal values on A and A' , but restricting it to the nonsaturated bad set destroys that property. Thus exact source localization and quotient invariance are in tension.

This does not disprove (EAT); it identifies what is missing. A proof would need an energy-sensitive, null-respecting carrier or a dual target-side cochain homotopy whose residual term is nevertheless controlled by the unsaturated exact-plaque mass in (10).

Remaining open issues

The document is therefore a conditional reduction, not a complete solution.

1. The essential mathematical gap is (EAT), an averaged tightening estimate for the degree-carrying part of the slices Z_y . It is used only in the last section, but it is indispensable for the high-aspect hard case.
2. The first alternative, the single-slice bound, the weighted coarea estimate, the essential filling lower bound, the exact-plaque Fubini estimate, and the absorption algebra are proved above.

3. Standard approaches tried so far do not close the gap: monomial degree estimates give only (1); ordinary source deformations create uncontrolled f -null boundary errors; quotient cochains are not compatible with nonsaturated exact-plaque localization as shown above.
4. A complete proof would require either a proof of (EAT), an equivalent dual formulation using only target-pullback cochains, or a different hard-regime argument exploiting the full two-parameter family of slices and the ambient mixed 2-Jacobian bounds.

References

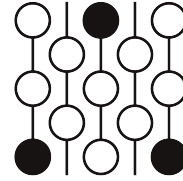
- [1] L. Guth, *Area-expanding embeddings of rectangles*, arXiv:0710.0403.
- [2] L. Guth, *Area-contracting maps between rectangles*, Ph.D. thesis, Massachusetts Institute of Technology, 2005.
- [3] L. Guth, *Directional isoperimetric inequalities and rational homotopy invariants*, arXiv:0802.3549.

Review of Submission 4A

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The argument solves a special case of the problem, (Proposition 1) showing that there is a c_2 such that if $R_1 R_2^2 \leq c_2 S_1 S_2 S_3$, then $R_3 R_4 \geq c_2 S_3 S_4$. This is a good start and seems to be correct. The argument goes on to show a weaker bound that's off by a factor of $(S_3/S_2)^{\frac{1}{10}}$. It tries to prove the full bound, but I don't think it makes much progress. Ultimately, it states an inequality (EAT) which would solve the problem if it were true, but it doesn't give any justification for why this inequality should be true or should be easier to prove.

Novelty. The first part of the argument (the proof of Proposition 1) uses an isoperimetric inequality of Guth to derive lower bounds on the mass of some fibers of F . This is similar to arguments in Guth's paper "Area-expanding embeddings of rectangles."

The second part of the argument ("A sufficient averaged hard-case estimate") introduces some new ideas (the integrals E_y and I and the set $\mathcal{B}_z$) but it doesn't get far enough to demonstrate that these ideas are useful or explain how they are relevant to the problem.

2 Recommendation

Reject. The argument applies known methods to prove some weaker bounds, but doesn't solve the problem. On the positive side, it gives some useful citations and it explains the amount of progress it made.

3 Submission with inline comments

Partial proof: the first alternative and the remaining hard estimate

Problem statement and interpretation

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The current proof is complete through the first alternative and through a conditional reduction of the hard case. It is not a complete proof of the stated dichotomy, because the averaged tightening estimate (EAT) below remains unproved. The arguments are written for smooth transverse maps; piecewise smooth maps may be treated by the same current-slicing argument, or by smoothing on strata with an arbitrarily small C^1 -loss and then absorbing that loss into the universal constants.

Standard inputs

All constants are positive and universal. If $\text{Dil}_2(f) \leq 1$, then $\text{Dil}_3(f), \text{Dil}_4(f) \leq 1$: for singular values $\sigma_1 \geq \dots \geq \sigma_4$, the inequality $\sigma_1 \sigma_2 \leq 1$ implies $\sigma_2 \leq 1$ and $\sigma_1 \dots \sigma_j \leq \sigma_1 \sigma_2$ for $j \geq 2$.

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Consequently

$$\text{Fill}_R(Z) \geq L \implies \text{Mass } Z \geq c \min\{R_1 R_2, L^{2/3}, (R_1 L)^{1/2}\}, \quad (2)$$

and, after decreasing c_* ,

$$\text{Mass } Z < c_* R_1 R_2 \implies \text{Fill}_R(Z) \leq C R_1 R_2^2. \quad (3)$$

The first alternative

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Proof. Choose a 1-Lipschitz function ψ on the (x_3, x_4) -rectangle which vanishes on its boundary and satisfies $\psi \geq c S_3$ on Q . Put $\beta = \psi dx_1 \wedge dx_2$. Then $d\beta$ has comass at most 1, and Stokes' theorem gives

$$\text{Mass } Y \geq \left| \int_Y d\beta \right| = \left| \int_{P_y} \beta \right| \geq c S_1 S_2 S_3.$$

The relative boundary terms vanish because $dx_1 \wedge dx_2 = 0$ on the x_1, x_2 -faces and $\psi = 0$ on the x_3, x_4 -faces. $\square$

Proposition 1. *There is $c_2 > 0$ such that, under the hypotheses of the problem, if*

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then $R_3 R_4 \geq c_2 S_3 S_4$.

Proof. Let $T = [[R]]$ be the fundamental relative current and put $F = (f_3, f_4)$. Since f has relative degree 1, $f_{\#}T = [[S]]$ as a relative 4-current. For a.e. $y \in Q$, slicing naturality gives

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Write $Z_y = \langle T, F, y \rangle$. Any relative filling of Z_y in R pushes forward, using $\text{Dil}_3(f) \leq 1$, to a filling of P_y in S . Lemma 1 therefore gives

$$\text{Fill}_R(Z_y) \geq c S_1 S_2 S_3.$$

If c_2 is sufficiently small, (3) forces $\text{Mass } Z_y \geq c_* R_1 R_2$. Since $J_2 F \leq \text{Dil}_2(f) \leq 1$, the coarea formula yields

$$\text{Vol}(R) \geq \int_Q \text{Mass } Z_y dy \geq c R_1 R_2 S_3 S_4.$$

Cancelling $R_1 R_2$ proves the proposition. □

Reviewer 2

This all seems reasonable so far and proves a potentially useful special case.

The unconditional partial hard-case bound

Assume $R_1 = \alpha S_1 \leq S_1$ and that the first conclusion fails with a small constant. Proposition 1 gives

$$R_1 R_2^2 \geq c S_1 S_2 S_3. \tag{4}$$

For the slices above, (2) with $L = c S_1 S_2 S_3$ gives

$$\text{Mass } Z_y \geq c \min\{R_1 R_2, (S_1 S_2 S_3)^{2/3}, (R_1 S_1 S_2 S_3)^{1/2}\}.$$

Using (4) and $R_1 = \alpha S_1$, every term in the minimum is at least $c\alpha^{1/2} S_1 (S_2 S_3)^{1/2}$. Hence

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On the other hand, (1) gives

$$\text{Vol}(R) \geq c\alpha^{-2} S_1 S_2 S_3 S_4. \tag{6}$$

Optimizing between (5) and (6) gives

$$\text{Vol}(R) \geq c S_1 S_2^{3/5} S_3^{7/5} S_4.$$

Thus the remaining gap is exactly the high-aspect factor $(S_3/S_2)^{1/10}$.

This section is very nice - it's clear about what's left to prove and how far off the current arguments are from the problem statement

A sufficient averaged hard-case estimate

Let $G = (f_1, f_2)$, keep $F = (f_3, f_4)$, and define

$$E_y = \int_{Z_y} \|dG|_{TZ_y}\|_{HS}^2 d\|Z_y\|, \quad I = \int_Q E_y dy, \quad q = |Q|.$$

At a regular point of F , let $K = \ker dF$, and let $a \geq b$ be the singular values of $dF|_{K^\perp}$; then $ab = J_2 F$. For any unit $w \in K$, the mixed two-vector $df(w) \wedge df(v)$, where v is a unit vector with $|dF(v)| = a$, has a component of norm $|dG(w)|a$. Since $\text{Dil}_2(f) \leq 1$, $|dG(w)|a \leq 1$, and hence $|dG(w)|^2 J_2 F \leq b/a \leq 1$. Summing over an orthonormal basis of K gives

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By coarea,

$$I \leq C \text{Vol}(R). \quad (7)$$

Let $\Omega = \{y \in Q : E_y \leq 4I/q\}$, so $|\Omega| \geq 3q/4$.

For real normal currents define

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This class is nonempty by filling the actual slice Z_y . If C is admissible, then $f_\# C$ fills P_y and $\text{Mass}(f_\# C) \leq \text{Mass } C$; Lemma 1 gives

$$\text{EFill}_f(y) \geq c S_1 S_2 S_3. \quad (8)$$

Therefore

$$A_\Omega := \int_\Omega \text{EFill}_f(y) dy \geq c S_1 S_2 S_3 q \geq c S_1 S_2 S_3^2 S_4. \quad (9)$$

For $z \in Q$, let $\mathcal{B}_z$ be the set of $x \in R$ such that z belongs to the F -image of at least one coordinate 2-plaque through x . For fixed x , the set of such z 's has measure at most $C R_3 R_4$, because there are six plaques, each has area at most $R_3 R_4$, and $J_2 F \leq 1$ on each plaque. Hence, for every normal 3-current B ,

$$\frac{1}{q} \int_Q \text{Mass}(B \llcorner \mathcal{B}_z) dz \leq C \frac{R_3 R_4}{S_3 S_4} \text{Mass } B. \quad (10)$$

This seems true, but unusual. I'd like to see some explanation of why this might be useful.

The following estimate is sufficient for the desired theorem. Take measurable near-minimizers C_y^0 with $\text{Mass } C_y^0 \leq 2 \text{EFill}_f(y)$; standard measurable selection in the separable flat topology, or equivalently outer integration with ε -minimizers, is enough for this formal step. One needs

$$A_\Omega \leq C \left(R_1 I + \frac{I^2}{S_1 q} \right) + \frac{C}{q} \int_Q \int_\Omega \text{Mass}(C_y^0 \llcorner \mathcal{B}_z) dy dz. \quad (\text{EAT})$$

Assuming (EAT), (10) and $\text{Mass } C_y^0 \leq 2 \text{ EFill}_f(y)$ give

$$A_\Omega \leq C \left(R_1 I + \frac{I^2}{S_1 q} \right) + C \frac{R_3 R_4}{S_3 S_4} A_\Omega.$$

If $R_3 R_4 \leq \kappa S_3 S_4$, the last term is absorbed by taking κ small. If also $R_1 \leq \kappa S_1$ and $I \leq \delta S_1 S_2^{1/2} S_3^{3/2} S_4$, then, since $q \sim S_3 S_4$,

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For small δ, κ this contradicts (9). Thus $I \geq c S_1 S_2^{1/2} S_3^{3/2} S_4$, and (7) gives the desired second alternative.

Reviewer 2

The argument reduces the original problem to proving (EAT), but it isn't clear that (EAT) is true or why to consider this particular inequality.

Why the remaining estimate is non-routine

The estimate (EAT) cannot be obtained by a standard source-local cellular deformation in the quotient by f -null chains without a new idea. The obstruction is already visible in a two-cell model. Let A, A' be two oriented 2-cells with $f_\# A = f_\# A' = P$. Then $N = A - A' \in \ker f_\#$. If an exact-plaque bad set contains A but not A' , the source-local projection P_{bad} with $P_{\text{bad}} A = A$, $P_{\text{bad}} A' = 0$ gives $f_\# P_{\text{bad}} N = P \neq 0$. Dually, a quotient cochain has equal values on A and A' , but restricting it to the nonsaturated bad set destroys that property. Thus exact source localization and quotient invariance are in tension.

Reviewer 2

This paragraph seems to be gesturing toward something, but it introduces a lot of terms (source-local, exact-plaque bad set, quotient cochain, exact source localization, quotient invariance) that the reader has to guess at. I couldn't extract much meaning from it.

This does not disprove (EAT); it identifies what is missing. A proof would need an energy-sensitive, null-respecting carrier or a dual target-side cochain homotopy whose residual term is nevertheless controlled by the unsaturated exact-plaque mass in (10).

Remaining open issues

The document is therefore a conditional reduction, not a complete solution.

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4. A complete proof would require either a proof of (EAT), an equivalent dual formulation using only target-pullback cochains, or a different hard-regime argument exploiting the full two-parameter family of slices and the ambient mixed 2-Jacobian bounds.

Reviewer 2

Again, several unexplained terms (absorption algebra, degree-carrying part, monomial degree estimate, nonsaturated exact-plaque localization). The gist seems to be that this is not a complete proof, but the details are fuzzy.

References

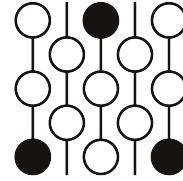
- [1] L. Guth, *Area-expanding embeddings of rectangles*, arXiv:0710.0403.
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- [3] L. Guth, *Directional isoperimetric inequalities and rational homotopy invariants*, arXiv:0802.3549.

Review of Submission 4A

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Editorial Note: This additional review includes comments that were volunteered by a reviewer. The editors did not ask the reviewer to reach a decision, given that the other reports were sufficient, but have decided to release this report as part of the public record in the belief that the reviewer comments can help further indicate where any errors lie.

Correctness.

Novelty.

2 Recommendation

3 Submission with inline comments

Partial proof: the first alternative and the remaining hard estimate

Problem statement and interpretation

I write the small universal constant in the problem as κ , reserving k for k -dilation. Rectangles are oriented Euclidean products and degree 1 means relative degree 1 on $H_4(R, \partial R; \mathbb{Z}) \rightarrow H_4(S, \partial S; \mathbb{Z})$.

The current proof is complete through the first alternative and through a conditional reduction of the hard case. It is not a complete proof of the stated dichotomy, because the averaged tightening estimate (EAT) below remains unproved. The arguments are written for smooth transverse maps; piecewise smooth maps may be treated by the same current-slicing argument, or by smoothing on strata with an arbitrarily small C^1 -loss and then absorbing that loss into the universal constants.

Standard inputs

All constants are positive and universal. If $\text{Dil}_2(f) \leq 1$, then $\text{Dil}_3(f), \text{Dil}_4(f) \leq 1$: for singular values $\sigma_1 \geq \cdots \geq \sigma_4$, the inequality $\sigma_1 \sigma_2 \leq 1$ implies $\sigma_2 \leq 1$ and $\sigma_1 \cdots \sigma_j \leq \sigma_1 \sigma_2$ for $j \geq 2$.

We use Guth's map-of-pairs estimate [1, Estimate 1]. For $n = 4, k = 2, j = 1, l = 4$ it gives

$$R_1^2 \text{Vol}(R) \geq c_0 S_1^3 S_2 S_3 S_4. \quad (1)$$

We also use the small-volume relative isoperimetric estimate in rectangles [1, Theorem 3]: if Z is a relative 2-cycle in R with $A = \text{Mass } Z \leq c_* R_1 R_2$, then

$$\text{Fill}_R(Z) \leq C_* \max\{A^{3/2}, A^2/R_1\}.$$

Consequently

$$\text{Fill}_R(Z) \geq L \implies \text{Mass } Z \geq c \min\{R_1 R_2, L^{2/3}, (R_1 L)^{1/2}\}, \quad (2)$$

and, after decreasing c_* ,

$$\text{Mass } Z < c_* R_1 R_2 \implies \text{Fill}_R(Z) \leq C R_1 R_2^2. \quad (3)$$

The first alternative

Let Q be the concentric subrectangle of the (x_3, x_4) -face of S with side lengths $S_3/2, S_4/2$, and set

$$P_y = [0, S_1] \times [0, S_2] \times \{y\}, \quad y \in Q.$$

Lemma 1 (central target plane). *Every relative 3-chain in S with boundary $P_y, y \in Q$, has mass at least $c S_1 S_2 S_3$.*

Proof. Choose a 1-Lipschitz function ψ on the (x_3, x_4) -rectangle which vanishes on its boundary and satisfies $\psi \geq c S_3$ on Q . Put $\beta = \psi dx_1 \wedge dx_2$. Then $d\beta$ has comass at most 1, and Stokes' theorem gives

$$\text{Mass } Y \geq \left| \int_Y d\beta \right| = \left| \int_{P_y} \beta \right| \geq c S_1 S_2 S_3.$$

The relative boundary terms vanish because $dx_1 \wedge dx_2 = 0$ on the x_1, x_2 -faces and $\psi = 0$ on the x_3, x_4 -faces. $\square$

Proposition 1. *There is $c_2 > 0$ such that, under the hypotheses of the problem, if*

$$R_1 R_2^2 \leq c_2 S_1 S_2 S_3,$$

then $R_3 R_4 \geq c_2 S_3 S_4$.

Proof. Let $T = [[R]]$ be the fundamental relative current and put $F = (f_3, f_4)$. Since f has relative degree 1, $f_{\#}T = [[S]]$ as a relative 4-current. For a.e. $y \in Q$, slicing naturality gives

$$f_{\#}\langle T, F, y \rangle = \langle [[S]], (x_3, x_4), y \rangle = \pm P_y.$$

Write $Z_y = \langle T, F, y \rangle$. Any relative filling of Z_y in R pushes forward, using $\text{Dil}_3(f) \leq 1$, to a filling of P_y in S . Lemma 1 therefore gives

$$\text{Fill}_R(Z_y) \geq c S_1 S_2 S_3.$$

If c_2 is sufficiently small, (3) forces $\text{Mass } Z_y \geq c_* R_1 R_2$. Since $J_2 F \leq \text{Dil}_2(f) \leq 1$, the coarea formula yields

$$\text{Vol}(R) \geq \int_Q \text{Mass } Z_y dy \geq c R_1 R_2 S_3 S_4.$$

Cancelling $R_1 R_2$ proves the proposition. $\square$

The unconditional partial hard-case bound

Assume $R_1 = \alpha S_1 \leq S_1$ and that the first conclusion fails with a small constant. Proposition 1 gives

$$R_1 R_2^2 \geq c S_1 S_2 S_3. \quad (4)$$

For the slices above, (2) with $L = c S_1 S_2 S_3$ gives

$$\text{Mass } Z_y \geq c \min\{R_1 R_2, (S_1 S_2 S_3)^{2/3}, (R_1 S_1 S_2 S_3)^{1/2}\}.$$

Using (4) and $R_1 = \alpha S_1$, every term in the minimum is at least $c \alpha^{1/2} S_1 (S_2 S_3)^{1/2}$. Hence

$$\text{Vol}(R) \geq c \alpha^{1/2} S_1 S_2^{1/2} S_3^{3/2} S_4. \quad (5)$$

On the other hand, (1) gives

$$\text{Vol}(R) \geq c \alpha^{-2} S_1 S_2 S_3 S_4. \quad (6)$$

Optimizing between (5) and (6) gives

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"Optimizing" to me is a strange description of what is happening. The step is correct but what's happening is raising inequality (5) to the 4th power, multiplying by inequality (6), and then taking the fifth root of the resulting inequality. It took me a little while to figure this out. The word "optimizing" suggests a more complicated process than this, perhaps taking a derivative or using Lagrange multipliers or using convexity/Jensen's inequality or something like that.

$$\text{Vol}(R) \geq c S_1 S_2^{3/5} S_3^{7/5} S_4.$$

Thus the remaining gap is exactly the high-aspect factor $(S_3/S_2)^{1/10}$.

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It is impressive to me that it was able to prove the previous inequality which is not so different from problem statement (up to this point everything seems correct), although the techniques were already in the literature.

A sufficient averaged hard-case estimate

Let $G = (f_1, f_2)$, keep $F = (f_3, f_4)$, and define

$$E_y = \int_{Z_y} \|dG|_{TZ_y}\|_{HS}^2 d\|Z_y\|, \quad I = \int_Q E_y dy, \quad q = |Q|.$$

At a regular point of F , let $K = \ker dF$, and let $a \geq b$ be the singular values of $dF|_{K^\perp}$; then $ab = J_2 F$. For any unit $w \in K$, the mixed two-vector $df(w) \wedge df(v)$, where v is a unit vector with $|dF(v)| = a$, has a component of norm $|dG(w)|a$. Since $\text{Dil}_2(f) \leq 1$, $|dG(w)|a \leq 1$, and hence $|dG(w)|^2 J_2 F \leq b/a \leq 1$. Summing over an orthonormal basis of K gives

$$\|dG|_{\ker dF}\|_{HS}^2 J_2 F \leq 2.$$

By coarea,

$$I \leq C \operatorname{Vol}(R). \quad (7)$$

Let $\Omega = \{y \in Q : E_y \leq 4I/q\}$, so $|\Omega| \geq 3q/4$.

For real normal currents define

$$\operatorname{EFill}_f(y) = \inf \{ \operatorname{Mass} C : C \in N_3(R, \partial R; \mathbb{R}), f_{\#}(\partial C) = \pm P_y \}.$$

This class is nonempty by filling the actual slice Z_y . If C is admissible, then $f_{\#}C$ fills P_y and $\operatorname{Mass}(f_{\#}C) \leq \operatorname{Mass} C$; Lemma 1 gives

$$\operatorname{EFill}_f(y) \geq cS_1S_2S_3. \quad (8)$$

Therefore

$$A_{\Omega} := \int_{\Omega} \operatorname{EFill}_f(y) dy \geq cS_1S_2S_3 q \geq cS_1S_2S_3^2S_4. \quad (9)$$

For $z \in Q$, let $\mathcal{B}_z$ be the set of $x \in R$ such that z belongs to the F -image of at least one coordinate 2-plaque through x . For fixed x , the set of such z 's has measure at most CR_3R_4 , because there are six plaques, each has area at most R_3R_4 , and $J_2F \leq 1$ on each plaque. Hence, for every normal 3-current B ,

$$\frac{1}{q} \int_Q \operatorname{Mass}(B \llcorner \mathcal{B}_z) dz \leq C \frac{R_3R_4}{S_3S_4} \operatorname{Mass} B. \quad (10)$$

The following estimate is sufficient for the desired theorem. Take measurable near-minimizers C_y^0 with $\operatorname{Mass} C_y^0 \leq 2 \operatorname{EFill}_f(y)$; standard measurable selection in the separable flat topology, or equivalently outer integration with ε -minimizers, is enough for this formal step. One needs

$$A_{\Omega} \leq C \left(R_1I + \frac{I^2}{S_1q} \right) + \frac{C}{q} \int_Q \int_{\Omega} \operatorname{Mass}(C_y^0 \llcorner \mathcal{B}_z) dy dz. \quad (\text{EAT})$$

Assuming (EAT), (10) and $\operatorname{Mass} C_y^0 \leq 2 \operatorname{EFill}_f(y)$ give

$$A_{\Omega} \leq C \left(R_1I + \frac{I^2}{S_1q} \right) + C \frac{R_3R_4}{S_3S_4} A_{\Omega}.$$

If $R_3R_4 \leq \kappa S_3S_4$, the last term is absorbed by taking κ small. If also $R_1 \leq \kappa S_1$ and $I \leq \delta S_1S_2^{1/2}S_3^{3/2}S_4$, then, since $q \sim S_3S_4$,

$$R_1I \leq C\kappa\delta S_1S_2S_3^2S_4, \quad \frac{I^2}{S_1q} \leq C\delta^2 S_1S_2S_3^2S_4.$$

For small δ, κ this contradicts (9). Thus $I \geq cS_1S_2^{1/2}S_3^{3/2}S_4$, and (7) gives the desired second alternative.

Why the remaining estimate is non-routine

The estimate (EAT) cannot be obtained by a standard source-local cellular deformation in the quotient by f -null chains without a new idea. The obstruction is already visible in a two-cell model. Let A, A' be two oriented 2-cells with $f_{\#}A = f_{\#}A' = P$. Then $N = A - A' \in \ker f_{\#}$. If an exact-plaque bad set contains A but not A' , the source-local projection P_{bad} with $P_{\text{bad}}A = A$, $P_{\text{bad}}A' = 0$ gives $f_{\#}P_{\text{bad}}N = P \neq 0$. Dually, a quotient cochain has equal values on A and A' , but restricting it to the nonsaturated bad set destroys that property. Thus exact source localization and quotient invariance are in tension.

This does not disprove (EAT); it identifies what is missing. A proof would need an energy-sensitive, null-respecting carrier or a dual target-side cochain homotopy whose residual term is nevertheless controlled by the unsaturated exact-plaque mass in (10).

Remaining open issues

The document is therefore a conditional reduction, not a complete solution.

1. The essential mathematical gap is (EAT), an averaged tightening estimate for the degree-carrying part of the slices Z_y . It is used only in the last section, but it is indispensable for the high-aspect hard case.
2. The first alternative, the single-slice bound, the weighted coarea estimate, the essential filling lower bound, the exact-plaque Fubini estimate, and the absorption algebra are proved above.
3. Standard approaches tried so far do not close the gap: monomial degree estimates give only (1); ordinary source deformations create uncontrolled f -null boundary errors; quotient cochains are not compatible with nonsaturated exact-plaque localization as shown above.
4. A complete proof would require either a proof of (EAT), an equivalent dual formulation using only target-pullback cochains, or a different hard-regime argument exploiting the full two-parameter family of slices and the ambient mixed 2-Jacobian bounds.

References

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